计算机网络实验报告

警示

1.实验报告如有雷同，雷同各方当次实验成绩均以 0 分计。
2.当次小组成员成绩只计学号、姓名登录在下表中的。
3.在规定时间内未上交实验报告的，不得以其他方式补交，当次成绩按 0 分计。
4.实验报告文件以 PDF 格式提交。

	计算机学院	班 级	计科 1 班		组长	陈安康
学号	22320021	22318001		22336070	22336064	
学生	陈安康	蔡可忻		冯乙山	杜雨彤	

【实验题目】生成树协议
【实验目的】理解快速生成树协议的配置及原理。使网络在有冗余链路的情况下避免环路的产生，避免广播风暴等。
【实验内容】

(1)完成实验教程实例 6-8 的实验，回答实验提出的问题及实验思考。(P204)
(2)抓取生成树协议数据包，分析桥协议数据单元（BPDU)。
(3)在实验设备上查看 VLAN 生成树，并学会查看其它相关重要信息。

【实验要求】

一些重要信息信息需给出截图。
注意实验步骤的前后对比！

本次实验完成后，请根据组员在实验中的贡献，请实事求是，自评在实验中应得的分数。(按百分制)

学号	学生	自评分
22320021	陈安康	100
22318001	蔡可忻	100
22336070	冯乙山	100
22336064	杜雨彤	100

【交实验报告】

上传实验报告：助教　　　　　　截止日期（不迟于）：1 周之内

上传包括两个文件：

（1）小组实验报告。上传文件名格式：小组号_生成树实验.pdf （由组长负责上传）

例如: 文件名“10_生成树实验.pdf”表示第 10 组的生成树实验报告

（2）小组成员实验体会。每个同学单独交一份只填写了实验体会的实验报告。只需填写自己的学号和姓名。

文件名格式：小组号_学号_姓名_生成树实验.pdf （由组员自行上传）

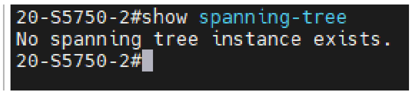
例如: 文件名“10_05373092_张三_生成树实验.pdf”表示第 10 组的生成树实验报告。

注意：不要打包上传！

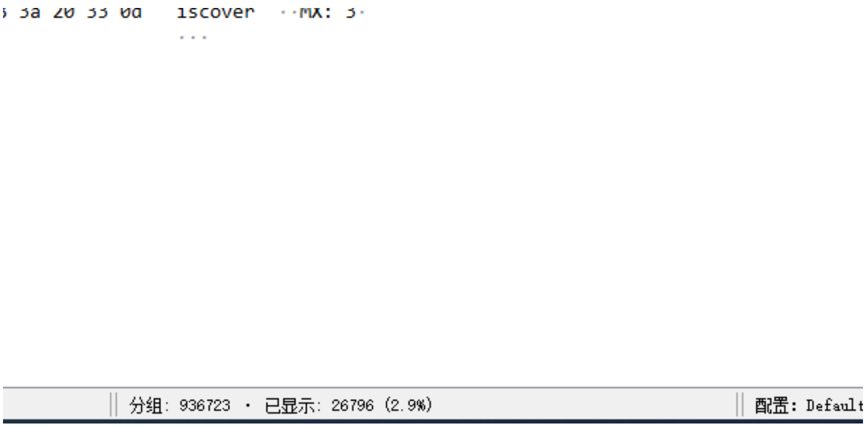
第（2）问见步骤 9 实验思考前 BPDU 分析

第（3）问命令见 show spanning-tree 相关命令，具体见实验流程

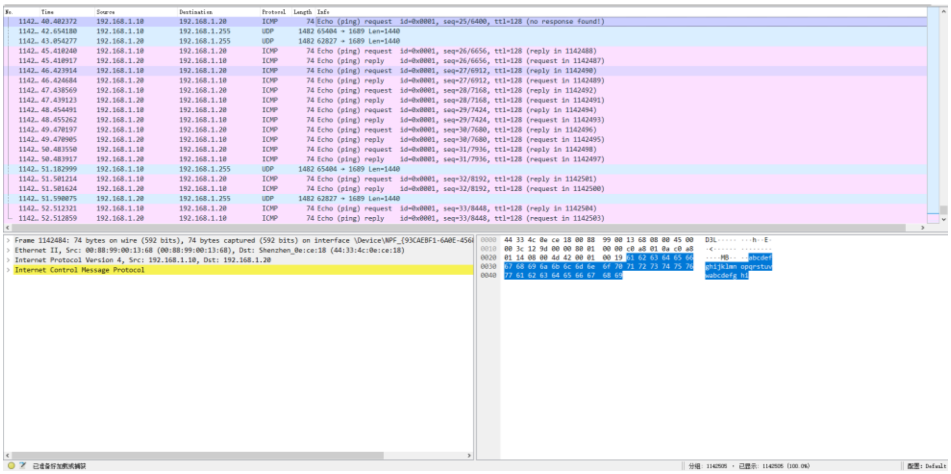
步骤 1：一开始的时候，没有生成树协议



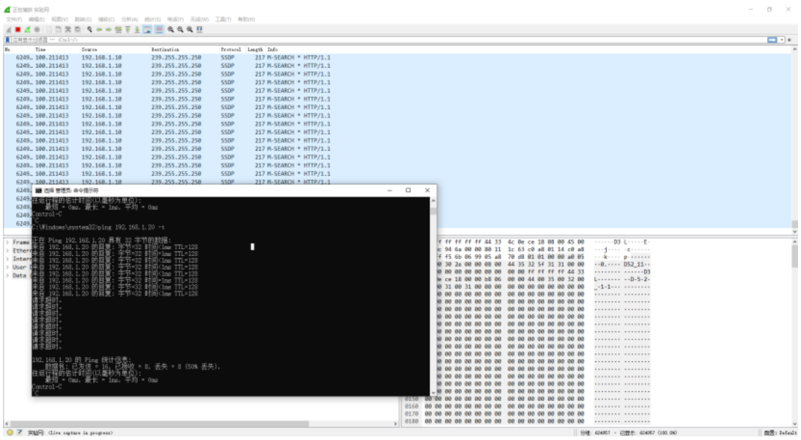
形成环路之后，产生网络风暴



（3）PC1 ping PC2

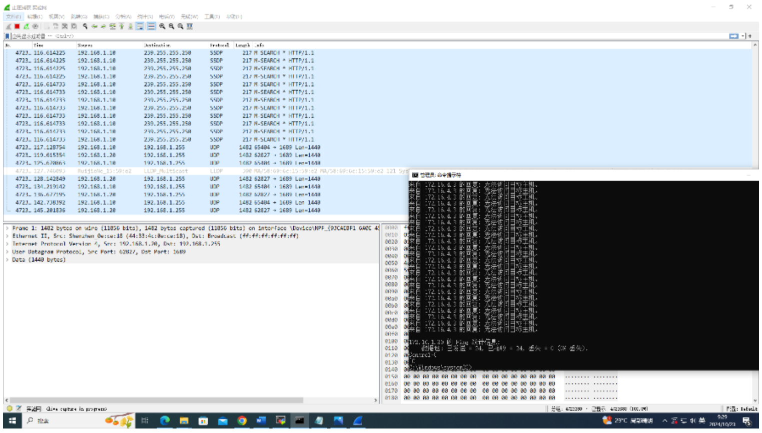


ping 一个不存在的 ip



情况 1 更快一些，ping 产生数据包会在 PC1 和 PC2 之间的环路中不断循环，而情况 2 因为没有真正 ping 到返回的包会少一些

此时终止 ping，广播风暴仍然存在，交换机没有启用环路检测机制，如生成树协议（STP）或快速生成树协议（RSTP），数据包依然在不断发送。



查看 mac 地址，发现在环路中，交换机会不断接收到相同的 MAC 地址，但来自不同的端口，会出现一个 MAC 地址对应多个端口的情况。

mac 地址对应的端口在不断发生变化

```
show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        0088.9900.1368       DYNAMIC  GigabitEthernet 0/1
   1        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
20-S5750-1#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        0088.9900.1368       DYNAMIC  GigabitEthernet 0/1
   1        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/2
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/2
20-S5750-1#*Oct 24 02:00:59: %LLDP-4-AGEOUTREM: Port GigabitEthernet 0/4 one neighbor aged out, Chassis ID is 5869.6c15.59ca, Port ID is Gi0/1.
show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        0088.9900.1368       DYNAMIC  GigabitEthernet 0/2
   1        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
20-S5750-1#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        0088.9900.1368       DYNAMIC  GigabitEthernet 0/1
   1        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/2
20-S5750-1#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        0088.9900.1368       DYNAMIC  GigabitEthernet 0/1
   1        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/2
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/2
20-S5750-1#*Oct 24 02:01:55: %LINK-3-UPDOWN: Interface GigabitEthernet 0/1, changed state to down.
*Oct 24 02:01:55: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/1, changed state to down.
```

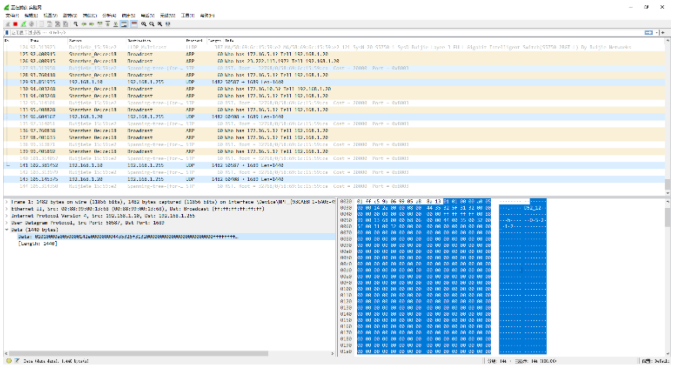
步骤 4：配置快速生成树协议，并用 show 查看

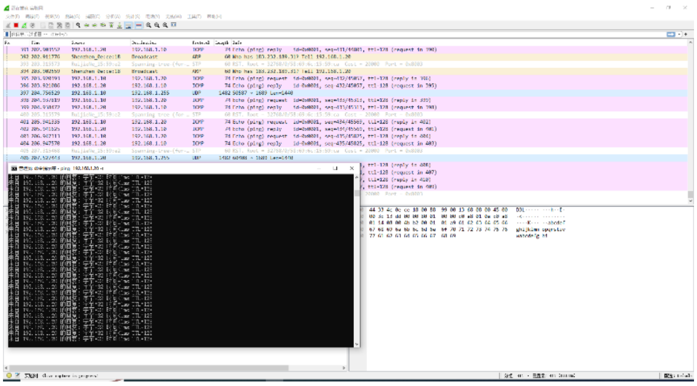
```
Enter configuration commands, one per line.  End with CNTL/Z.
20-S5750-1(config)#vlan 10
20-S5750-1(config-vlan)#name sales
20-S5750-1(config-vlan)#exit
20-S5750-1(config)#interface gigabitethernet 0/3*Oct 24 02:06:10: %LLDP-4-AGEOUTREM: P
ort GigabitEthernet 0/2 one neighbor aged out, Chassis ID is 5869.6c15.59ca, Port ID i
s Gi0/2.

20-S5750-1(config-if-GigabitEthernet 0/3)#switchport access vlan 10
20-S5750-1(config-if-GigabitEthernet 0/3)#exit
20-S5750-1(config)#interface range gigabitethernet 0/1-2
20-S5750-1(config-if-range)#switchport mode trunk
20-S5750-1(config-if-range)#exit
20-S5750-1(config)#spanning-tree
Enable spanning-tree.
20-S5750-1(config)#spanning-tree mode rstp
20-S5750-1(config)#*Oct 24 02:10:32: %SPANTREE-6-PORTFASTCHG: Port GigabitEthernet 0/1
 portfast state changed from edge to non-edge.
*Oct 24 02:10:32: %SPANTREE-5-ROOTCHANGE: Root Changed: New Root Port is GigabitEthern
et 0/1. New Root Mac Address is 5869.6c15.59ca.
*Oct 24 02:10:34: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/1
 on MST0.
*Oct 24 02:10:59: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/1
 on MST0.

20-S5750-1(config)#show spanning-tree
StpVersion : RSTP
SysStpStatus : ENABLED
MaxAge : 20
HelloTime : 2
ForwardDelay : 15
BridgeMaxAge : 20
BridgeHelloTime : 2
BridgeForwardDelay : 15
MaxHops: 20
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef  : Disabled
BridgeAddr : 5869.6c15.59e2
Priority: 32768
TimeSinceTopologyChange : 0d:0h:1m:45s
TopologyChanges : 1
DesignatedRoot : 32768.5869.6c15.59ca
RootCost : 20000
RootPort : GigabitEthernet 0/1
20-S5750-1(config)#*Oct 24 02:12:25: %LINK-3-UPDOWN: Interface GigabitEthernet 0/2, ch
anged state to up.
*Oct 24 02:12:25: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/2,
 changed state to up.
```

再次连线形成物理环路，发现无论发包与否都没有网络风暴产生





查看 mac 地址

```
20-S5750-1(config)#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
  10        0088.9900.1368       DYNAMIC  GigabitEthernet 0/3
  10        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
20-S5750-1(config)#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
  10        0088.9900.1368       DYNAMIC  GigabitEthernet 0/3
  10        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
20-S5750-1(config)#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
  10        0088.9900.1368       DYNAMIC  GigabitEthernet 0/3
  10        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
20-S5750-1(config)#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
  10        0088.9900.1368       DYNAMIC  GigabitEthernet 0/3
  10        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
20-S5750-1(config)#show mac-address-table
Vlan        MAC Address          Type     Interface
----------  -------------------- -------- -------------------
   1        5869.6c15.59ca       DYNAMIC  GigabitEthernet 0/1
  10        0088.9900.1368       DYNAMIC  GigabitEthernet 0/3
  10        4433.4c0e.ce18       DYNAMIC  GigabitEthernet 0/1
20-S5750-1(config)#
```

Mac 地址对应端口稳定不变化

步骤 5：查看端口状态

```
DesignatedRoot : 32768.5869.6c15.59ca
RootCost : 20000
RootPort : GigabitEthernet 0/1
20-S5750-1(config)#show spanning-tree interface gigabitethernet 0/1

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 32768.5869.6c15.59ca
PortDesignatedCost : 0
PortDesignatedBridge :32768.5869.6c15.59ca
PortDesignatedPortPriority : 128
PortDesignatedPort : 1
PortForwardTransitions : 2
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-1(config)#show spanning-tree interface gigabitethernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 32768.5869.6c15.59ca
PortDesignatedCost : 0
PortDesignatedBridge :32768.5869.6c15.59ca
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 0
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : alternatePort
20-S5750-1(config)#
20-S5750-1#*Oct 24 02:20:08: %SYS-5-CONFIG_I: Configured from console by console
```

```
2. 172.16.4.5 2001    3. 172.16.4.5 2002
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef  : Disabled
BridgeAddr : 5869.6c15.59ca
Priority: 32768
TimeSinceTopologyChange : 0d:0h:7m:18s
TopologyChanges : 2
DesignatedRoot : 32768.5869.6c15.59ca
RootCost : 0
RootPort : 0
20-S5750-2(config)#show spanning-tree interface gigabitethernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Enabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Enabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 32768.5869.6c15.59ca
PortDesignatedCost : 0
PortDesignatedBridge :32768.5869.6c15.59ca
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 1
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : designatedPort
20-S5750-2(config)#show spanning-tree interface gigabitethernet 0/1

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 32768.5869.6c15.59ca
PortDesignatedCost : 0
PortDesignatedBridge :32768.5869.6c15.59ca
PortDesignatedPortPriority : 128
PortDesignatedPort : 1
PortForwardTransitions : 2
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : designatedPort
20-S5750-2(config)#
```


交换机 A 的 0/1 端口处于 forwarding，0/2 端口 discarding

交换机 B 的 0/1 端口 forwarding，0/2 端口 forwarding

所以交换机 B 为根交换机

步骤 6：设置交换机优先级并查看

```
20-S5750-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
20-S5750-1(config)#spanning-tree priority 4096
20-S5750-1(config)#*Oct 24 02:24:58: %SPANTREE-5-EVENT: The device has been selected a
s the Root Bridge.
*Oct 24 02:24:58: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2
 on MST0.
*Oct 24 02:24:58: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/1
 on MST0.
*Oct 24 02:24:58: %SPANTREE-6-RX_INFBPDU: Received inferior BPDU on port GigabitEthern
et 0/1.
*Oct 24 02:24:58: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/1
 on MST0.
*Oct 24 02:24:58: %SPANTREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 02:24:59: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/1
 on MST0.
*Oct 24 02:25:00: %SPANTREE-5-TOPOTRAP: Topology Change Trap.

20-S5750-1(config)#show spanning-tree
StpVersion : RSTP
SysStpStatus : ENABLED
MaxAge : 20
HelloTime : 2
ForwardDelay : 15
BridgeMaxAge : 20
BridgeHelloTime : 2
BridgeForwardDelay : 15
MaxHops: 20
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef  : Disabled
BridgeAddr : 5869.6c15.59e2
Priority: 4096
TimeSinceTopologyChange : 0d:0h:0m:14s
TopologyChanges : 2
DesignatedRoot : 4096.5869.6c15.59e2
RootCost : 0
RootPort : 0
20-S5750-1(config)#
```

查看交换机生成树配置信息

设置优先级之后，交换机 A 变为根交换机

```
RootPort : GigabitEthernet 0/1
20-S5750-2(config)#show spanning-tree interface gigabitethernet 0/1

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 1
PortForwardTransitions : 2
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#show spanning-tree interface gigabitethernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 1
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : alternatePort
20-S5750-2(config)#
```

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交换机 B 0/1 变为 forwarding 状态，0/2 为 discarding，0/1 为根端口

（1）由以上信息可以得知表格如下，其中交换机A作为根交换机。

col1	交换机A	交换机B
Priority	4096	32768
BridgeAddr	5869.6c15.59e2	5869.6c15.59ca
DesignatedRoot	4096.5869.6c15.59e2	4096.5869.6c15.59e2
RootCost	0	20000
RootPort	-	0/1
Designated	0/1 0/2	-

（2）此时若将端口0/1之间的链路down掉，则交换机B的0/2端口由阻塞状态转变为转发状态，首先在拔掉之前运行命令查看交换机B的0/2端口状态，此时仍然是discarding，在切断端口0/1链路后，经过2秒时间后（计时器计时）再次查看交换机B的0/2端口状态，发现已经改变变为forwarding，整体用时2s左右：

```
PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 7
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : alternatePort
20-S5750-2(config)#*Oct 24 03:03:20: %SPANTREE-5-ROOTCHANGE: Root Changed: New Root Po
rt is GigabitEthernet 0/2. New Root Mac Address is 5869.6c15.59e2.
*Oct 24 03:03:21: %SPANTREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 03:03:22: %LINK-3-UPDOWN: Interface GigabitEthernet 0/1, changed state to down
.
*Oct 24 03:03:22: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/1,
 changed state to down.
show spanning-tree interface gigabitEthernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 8
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#
```

（3）此时交换机的信息如下：

交换机A:

```
20-S5750-1(config)#show spanning-tree
StpVersion : RSTP
SysStpStatus : ENABLED
MaxAge : 20
HelloTime : 2
ForwardDelay : 15
BridgeMaxAge : 20
BridgeHelloTime : 2
BridgeForwardDelay : 15
MaxHops: 20
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef  : Disabled
BridgeAddr : 5869.6c15.59e2
Priority: 4096
TimeSinceTopologyChange : 0d:0h:2m:5s
TopologyChanges : 3
DesignatedRoot : 4096.5869.6c15.59e2
RootCost : 0
RootPort : 0
20-S5750-1(config)#
```

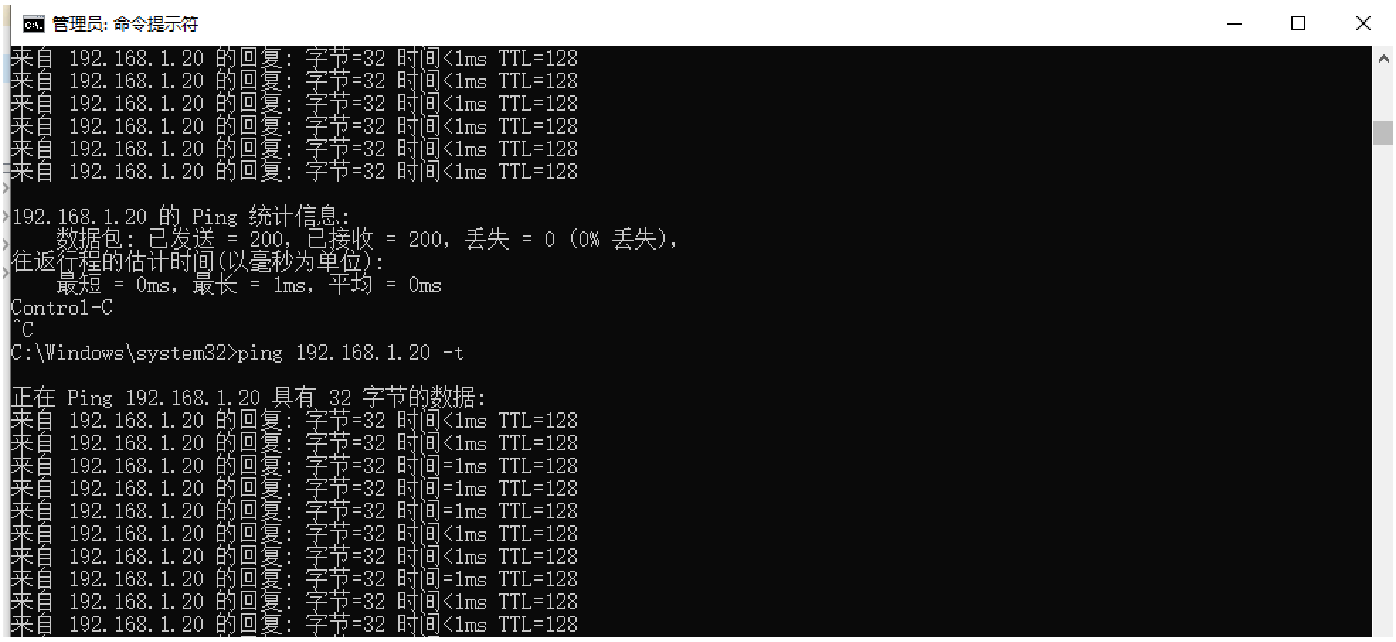
交换机B：

```
20-S5750-2(config)#show spanning-tree
StpVersion : RSTP
SysStpStatus : ENABLED
MaxAge : 20
HelloTime : 2
ForwardDelay : 15
BridgeMaxAge : 20
BridgeHelloTime : 2
BridgeForwardDelay : 15
MaxHops: 20
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef  : Disabled
BridgeAddr : 5869.6c15.59ca
Priority: 32768
TimeSinceTopologyChange : 0d:0h:2m:44s
TopologyChanges : 4
DesignatedRoot : 4096.5869.6c15.59e2
RootCost : 20000
RootPort : GigabitEthernet 0/2
20-S5750-2(config)#
```

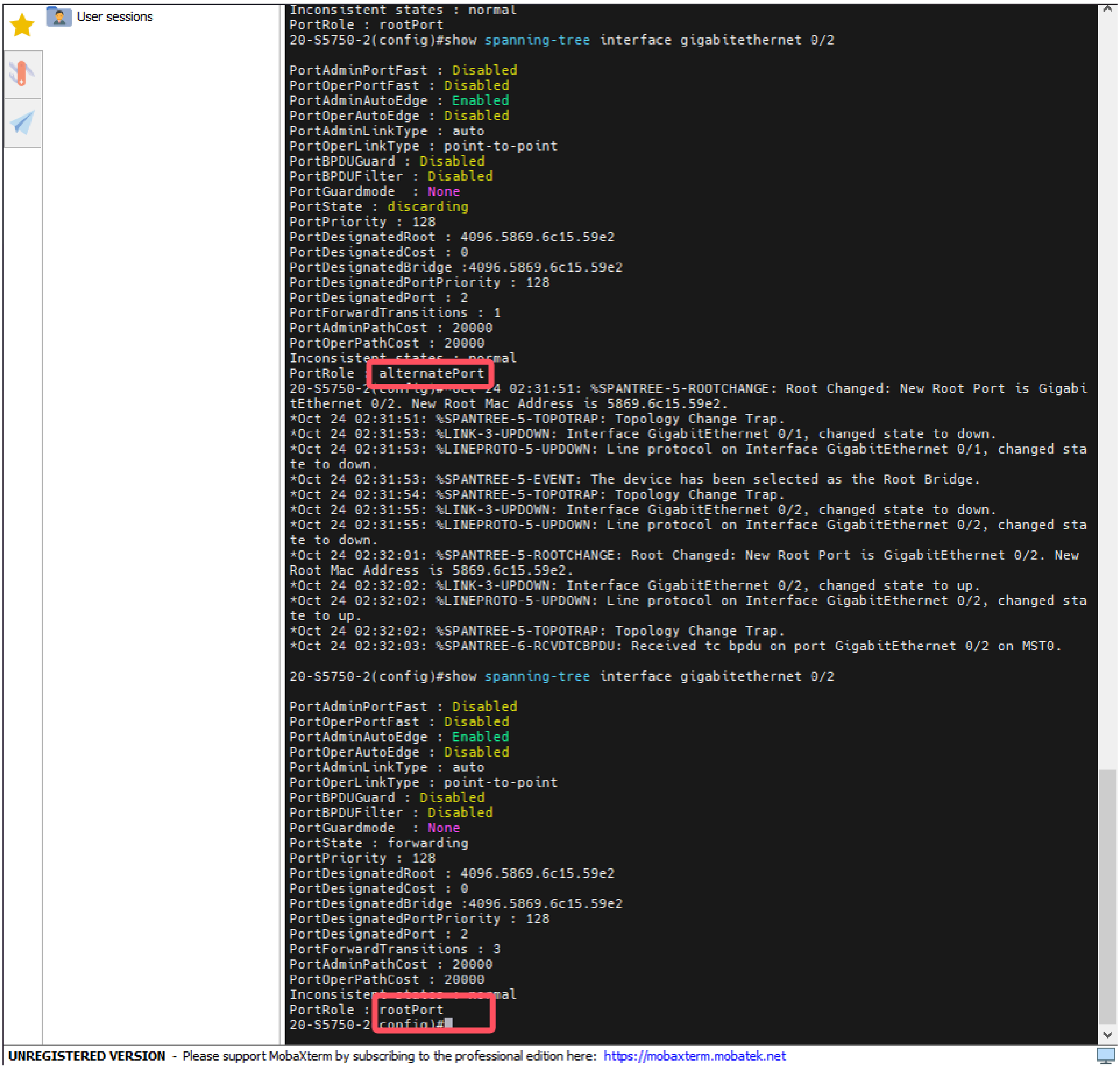
col1	交换机A	交换机B
Priority	4096	32768
BridgeAddr	5869.6c15.59e2	5869.6c15.59ca
DesignatedRoot	4096.5869.6c15.59e2	4096.5869.6c15.59e2
RootCost	0	20000
RootPort	-	0/2
Designated	0/1 0/2	-

与（1）进行比较，可以发现交换机B的根端口已经转变为了端口0/2，说明在端口0/1链路断开后，交换机会重新计算生成树，动态调整使得端口0/2处于打开状态，确保新的拓扑结构能够稳定下来，防止环路。

（4）将网络拓扑恢复到没有断开端口0/1之前的状态，此时交换机B是非根交换机，端口0/1是根端口，端口0/2是阻塞端口。然后执行ping命令，并在过程中拔掉交换机A与交换机B端口0/1之间的连线，发现并没有丢包产生：



过程中交换机的输出如下，从图中可以观察到，在这个过程中，端口0/2从阻塞端口（替换端口）变为了根端口：



正是端口0/2从替换端口转换为了根端口，承担起了数据转发的任务，才使得网络连接通常，没有丢包现象的发生。

（5）记录表格如下：

col1	交换机A	交换机B
Priority	4096	32768
BridgeAddr	5869.6c15.59e2	5869.6c15.59ca
DesignatedRoot	4096.5869.6c15.59e2	4096.5869.6c15.59e2
RootCost	0	20000
RootPort	-	0/2
Alternate	-	-

同（1）表格进行比较可以发现，交换机B的根端口变成了端口0/2，原因分析同（3）。

注意，由于交换机B上的端口0/1是断开连接了，因此它不能起到替换端口在必要时迅速切换为转发状态的端口的作用，而是被交换机直接关闭了，因此它不属于交换端口，交换端口必须要求端口是连接状态，只是被阻塞了而已。

启动监控软件 Wireshark，捕获 BPDU，并进行协议分析。

A.

```
> Frame 857: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on inte
> IEEE 802.3 Ethernet
> Logical-Link Control
v Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Rapid Spanning Tree (2)
    BPDU Type: Rapid/Multiple Spanning Tree (0x02)
  > BPDU flags: 0x7c, Agreement, Forwarding, Learning, Port Role: Designated
  v Root Identifier: 32768 / 0 / 58:69:6c:15:59:ca
      Root Bridge Priority: 32768
      Root Bridge System ID Extension: 0
      Root Bridge System ID: RuijieNetwor_15:59:ca (58:69:6c:15:59:ca)
    Root Path Cost: 20000
  v Bridge Identifier: 32768 / 0 / 58:69:6c:15:59:e2
      Bridge Priority: 32768
      Bridge System ID Extension: 0
      Bridge System ID: RuijieNetwor_15:59:e2 (58:69:6c:15:59:e2)
    Port identifier: 0x8003
```

经过 tcp 过滤器之后，可以观察到:

Bridge Identifier: 32768/0/58:69:6c:15:59:e2

Bridge Priority: 32768

Root Path Cost: 20000

Port identifier: 0x8003

B.

```
Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Rapid Spanning Tree (2)
    BPDU Type: Rapid/Multiple Spanning Tree (0x02)
  > BPDU flags: 0x7c, Agreement, Forwarding, Learning, Port Role: Designated
  v Root Identifier: 4096 / 0 / 58:69:6c:15:59:e2
      Root Bridge Priority: 4096
      Root Bridge System ID Extension: 0
      Root Bridge System ID: RuijieNetwor_15:59:e2 (58:69:6c:15:59:e2)
    Root Path Cost: 0
  v Bridge Identifier: 4096 / 0 / 58:69:6c:15:59:e2
      Bridge Priority: 4096
      Bridge System ID Extension: 0
      Bridge System ID: RuijieNetwor_15:59:e2 (58:69:6c:15:59:e2)
    Port identifier: 0x8003
```

可以观察到:

Bridge Identifier: 4096 / 0 / 58:69:6c:15:59:e2

Root Bridge Priority: 4096

Root Path Cost: 0

Port identifier: 0x8003

【实验思考】

（1）请问该实验中有无环路？请说明判断的理由。如果存在，说明交换机是如何避免环路的？

有出现环路，因为出现了“广播风暴”现象，交换机利用生成树算法，在有交换环路的网络中生成一个没有环路的树形网络。运用该算法将交换网络冗余的备份链路在

逻辑上断开，当主要链路出现故障时，能够自动切换到备份链路以保证数据的正常转发。

（2） 冗余链路会不会出现 MAC 地址表不稳定和多帧复制的问题？请举例说明。

会出现该问题。在具有冗余链路的网络中，交换机可能接收到来自同一设备的多个不同源 MAC 地址的帧，从而导致 MAC 地址表不稳定。在具有冗余链路的网络中，同一帧在网络中被多次转发给目标设备还可能导致多帧复制的问题。

以实验中的拓扑图为例，如果没有 STP 协议，PC1 发送数据帧给交换机 A，沿着 F0/1 发送到交换机 B，然后由交换机 B 又以 F0/2 端口发送到交换机 A。这样反复执行，交换机 A 和 B 在 F0/1 和 F0/2 反复接收 PC1 的 MAC 地址，使得 MAC 地址表不稳定。

同样，PC1 发送一个数据包到 PC2，有 F0/1 和 F0/2 两条路可以走，所以两条路都发送了。然后 PC2 就会接到两个（多个）数据帧。

（3） 将实验改用 STP 协议，重点观察状态转换时间。

```
PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 6
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : alternatePort
20-S5750-2(config)#*Oct 24 02:59:58: %SPANTREE-5-ROOTCHANGE: Root Changed: New Root Port is Gigabi
tEthernet 0/2. New Root Mac Address is 5869.6c15.59e2.
*Oct 24 03:00:00: %LINK-3-UPDOWN: Interface GigabitEthernet 0/1, changed state to down.
*Oct 24 03:00:00: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/1, changed sta
te to down.
show spanning-tree interface gigabitEthernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 6
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
```

```
20-S5750-2(config)#show spanning-tree interface gigabitEthernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : learning
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 6
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#*Oct 24 03:00:28: %SPANTREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 03:00:29: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:31: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:33: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
show spanning-tree interface gigabitEthernet 0/2*Oct 24 03:00:35: %SPANTREE-6-RCVDTCBPDU: Received
 tc bpdu on port GigabitEthernet 0/2 on MST0.


PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode  : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge :4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 7
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#*Oct 24 03:00:37: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEther
net 0/2 on MST0.
*Oct 24 03:00:39: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:41: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:43: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:45: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:47: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:49: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
*Oct 24 03:00:51: %SPANTREE-6-RCVDTCBPDU: Received tc bpdu on port GigabitEthernet 0/2 on MST0.
```

状态转换时间大约为：46s

（4）在本实验中，开始时首先在两台交换机之间只连接一根跳线，发现可以正常 ping 通。此时在两台交换机之间多接一根跳线，发现还是可以继续正常 ping 通。请问此时有广播风暴吗？

没有广播风暴。